

React DevCamp Handbook

React DevCamp

Overview

Welcome, aspiring React developers! According to Stack Overflow's 2023 survey, React is the most popular enterprise-grade web technology in the world — which makes it an invaluable tool to have on your toolbox! This DevCamp is aimed at sharpening your skills, expanding your understanding, and empowering you to create captivating and functional React web applications. Over the next few weeks, you'll have the chance to apply fundamental React concepts and techniques to real-world scenarios, gaining practical experience that will kickstart your journey as a React developer.

Duration

- 8 Weeks

Prerequisites

- JavaScript Basics
- NodeJS Basics

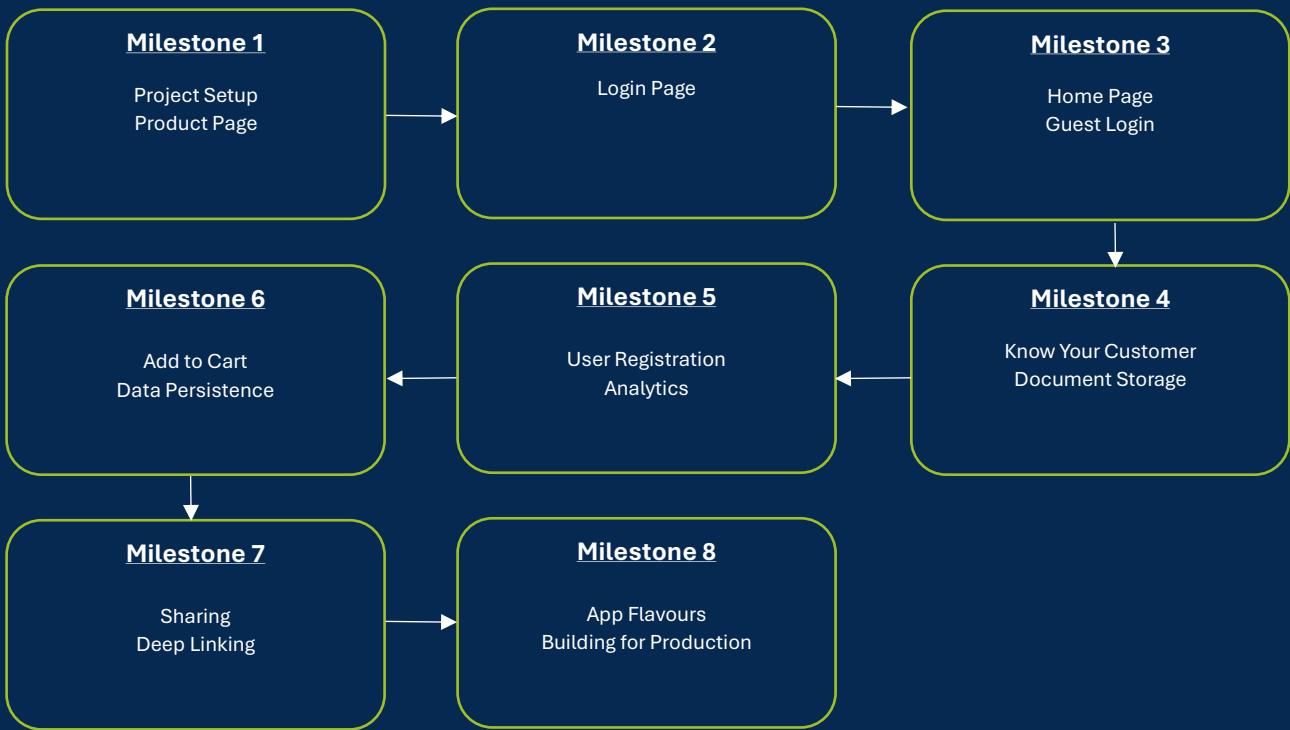
Technical Requirements

- [NodeJS and NPM](#)
- [VS Code](#) or [WebStorm](#)
- [Git](#)

Concepts Covered

- React Basics
- React Project Scaffolding
- React Code Quality & Linting
- React Components & Props
- React State & Hooks
- React Routing
- React Forms
- React Component Libraries
- Custom React Hooks
- React Context
- Building React For Production
- React Unit Testing

Timeline



Getting Started

Setting up your development environment

Before we dive into the exciting world of React development, let's ensure your development environment is set up correctly. Follow these steps to get everything in order:

1. Install NodeJS

React uses NodeJS to build and bundle its JavaScript runtime code. Please download the latest LTS version of NodeJS [here](#). Installing NodeJS will also install NPM (Node Package Manager), which React (and all NodeJS projects) uses to manage its dependencies.

You can verify that NodeJS and NPM have installed correctly by running the following commands from any terminal.

- `node -v`
- `npm -v`

If NodeJS and NPM are correctly installed, these commands should list the installed versions, respectively:

```
PS C:\> node -v
v20.11.0
PS C:\> npm -v
10.2.4
```

If you receive an error stating that either NodeJS or NPM is not an internal or external command, make sure that NodeJS and NPM have been added to your system path variable. This should be done for you automatically during the installation process, but in case this did not happen you can follow [this guide](#).

2. Install IDE

For JavaScript and TypeScript development, WebStorm and VS Code both offer great development experiences. WebStorm is paid-for, but already comes bundled with all the necessary React development tools out of the box. WebStorm does have a trial period if you'd like to try it out for free. VS Code is free but requires you to find and install your own framework related plugins. Its out-of-box React development experience is not great, but you can reach out to the Entelect [#react](#) slack channel to ask for some recommended React plugins to make your experience better.

WebStorm is built by JetBrains, so if you're used to their IDEs (such as IntelliJ and Rider), it will give you a more familiar development experience. VS Code is built by Microsoft, so if you're familiar with Microsoft Visual Studio, VS Code will be the more familiar option.

3. Set up Git

We'll be using Git for version control. If you don't have Git installed, follow the instructions on [the official Git website](#) to do so. Both WebStorm and VS Code have built-in Git integrations, which greatly improve your development experience; but you're welcome to use a different Git GUI tool if you prefer.

Next Steps

You should now be equipped with the essential tools and resources needed for the React DevCamp. Let's dive right in!

This guide is organised into weekly deliverables, each focussing React-specific development concepts and tasks. Start with Milestone 1 and progress through each week sequentially.

You'll find more details on the next page.

Need Help?

If you encounter any challenges or have questions, reach out to your Team Lead! Join the DevCamp Teams channel, where you connect with fellow engineers for support.

Now, let the coding adventure begin! 🧙 Happy coding and enjoy the journey of becoming a React developer.

Milestone Breakdown

Milestone 1

Project Setup
Product Page

Milestone 2

Login Page

Milestone 3

Home Page
Guest Login

Milestone 1

Deliverables

- **Pick A React Flavour**

When using React on an enterprise-grade system, you'll more than likely use one of React's recommended frameworks such as NextJS, Remix, Gatsby, or Expo (for native apps). However, if your main goal is to learn the core principles of React, it is recommended to use React without a framework. Once you understand React, its frameworks are very easy to pick up. You can learn more about React and its frameworks [here](#).

For this DevCamp, we'll be using Vite to scaffold our frameworkless React project for us. You can use either JavaScript or TypeScript with React. We recommend sticking with JavaScript to focus on the React learnings, but if you are experienced with TypeScript, you're welcome to use it. Have a look at Vite's [scaffolding documentation](#) to see how you can set up a project using either the **react** or **react-ts** templates.

- **Project Setup**

Create a brand-new frameworkless React project and push it to your Git repository. Give the project any name; make it creative if you'd like! (Refer to Appendix A to see what the app is about for inspiration).

- **Implement First Screen**

Build the product info page according to the designs and use it as your initial screen (the Figma design will state that it's for Android; however, you can build a mobile-first web-app version of these designs).

Resources

- BRS requirement (Appendix A)

US4 – View products

- Designs

Figma → Ready for Development → Home → [Product Page](#)

Useful Links

1. Getting Started with React

- [Quick Start](#)
- [Tutorial: Tic-Tac-Toe](#)
- [Thinking in React](#)

3. React Hooks

- [useState](#)
- [useEffect](#)

2. Learn React

- [Describing the UI](#)
- [Adding Interactivity](#)
- [Managing State](#)
 - Stop before "Extracting State Logic into a Reducer" subsection

4. Scaffolding React

- [Scaffolding Your First Project](#)
- [Editor Setup \(ESLint + Prettier\)](#)
- [React Developer Tools](#)

Milestone Breakdown



Milestone 2

Deliverables

- **Component Library**

When it comes to reusable components such as buttons, inputs, cards, etc; there's no need to reinvent the wheel. There are a ton of fantastic component and styling libraries out there for React, and you're welcome to try some of them out. Our recommendations are listed under the Useful Links section.

- **Form Library**

Just as with reusable components, there's no need to build form logic from scratch. There are many feature-rich form libraries for React, which you're welcome to use on your project. Our recommendations are listed under the Useful Links section.

- **Routing**

Since we'll be building more pages soon, we'll need a routing library for our application. [React Router](#) is a well-tested and mature routing library for React, which we'll be using it to facilitate routing within our applications.

- **Login Page**

Build a functional, user-friendly, and secure Login page. Ensure you use the provided auth service.

Resources

- BRS Requirements & Service (Appendix A)
US1 – Log in
8.1 Authentication Service

- Designs
Figma → Ready for Development → Log in → [Login Flow](#)

Useful Links

Component Libraries

- [Tailwind CSS](#)
- [React Semantic UI](#)
- [React Flowbite](#)

Routing Library

- [React Router Tutorial](#)

Form Libraries

- [Controlled And Uncontrolled Components](#)
- [React Hook Form](#) (recommended)
- [Formik](#)

Request Libraries

- [Axios](#) (simpler, not React specific)
- [React Query](#) (more powerful)

Milestone Breakdown



Milestone 3

Deliverables

- **Custom Hooks**

With React, it is best practice to reuse components to compose your application. You'll also find it helpful to make your application/business logic reusable. In many cases, it make sense to separate your render-logic from your business/application-logic. To do this, React allows you to create custom hooks. See the Useful Links section for more information on this.

- **Home Page**

Build the Home page to display available products to the user. Ensure you cater for loading and error states for your screens. Use the provided Catalogue service.

- **Guest Login**

Allow anyone to access the app as a guest. Only certain products are available; and some functionality is blocked. Ensure you look over the designs and BRS to understand the different login views.

Resources

- BRS Requirement & Service (Appendix A)

US4 – View Products

9.1.1 Product Catalogue

- Designs

Figma → Ready for Development → Home → [Guest Flow](#)

Useful Links

Advanced React Hooks

- [useCallback](#)
- [useMemo](#)
- [Custom Hooks](#)

Advanced State Management

- [React Context](#)
- [You probably don't need Redux](#)

React Hook Libraries

- [React Use](#)

Milestone Breakdown

Milestone 3

Home Page
Guest Login

Milestone 4

Know Your Customer
Document Storage

Milestone 5

User Registration
Analytics

Milestone 4

Deliverables

- **Know Your Customer (KYC)**

Implement the KYC flow where the user must upload proof of residence and a selfie. Allow the user to take a photo or select a file.

- **Document Storage**

Use Firebase to store the uploaded documents.

Resources

- BRS Requirement
US7 – Do Customer Checks

- Designs
Figma → Ready for Development → Registration → [KYC Flows](#)

Useful Links

Unit Testing React

- [Jest](#)
- [React Testing Library](#)

Firebase

- [Firebase For Web](#)

File Picker Libraries

- [Use File Picker](#)
- [Upload Widget React](#)

Webcam Library

- [React Webcam](#)

Milestone Breakdown

Milestone 4

Know Your Customer
Document Storage

Milestone 5

User Registration
Analytics

Milestone 6

Add to Cart
Data Persistence

Milestone 5

Deliverables

- **User Registration**

Build a user registration flow with input validation. Include the completed KYC flow as a step.

- **Analytics**

Use Firebase to add analytics to the app. Track when a user drops out of registration.

Resources

- **BRS Requirements & Service**

US2 – Create Profile
8.6.3 Register Customer

- **Designs**

Figma → Ready for Development → Registration → [Registration Flow](#)

Useful Links

Firestore

- [Firestore](#)

Firebase

- [Firebase Analytics For Web](#)

Milestone Breakdown

Milestone 5

User Registration
Analytics

Milestone 6

Add to Cart
Data Persistence

Milestone 7

Sharing
Deep Linking

Milestone 6

Deliverables

- **Add to Cart**

Allow the user to add items to their cart and check out. Users need to be logged in to complete the process.

- **Data Persistence**

When a user adds items to their cart, those items should be there after closing and reopening the website.

Resources

- **BRS Requirement**

US8 – Generate Contract

- **Designs**

Figma → Ready for Development → [Checkout](#)

Useful Links

Data Persistence

- [Using storage in React](#)
- [Browser Local Storage](#)

React Storage Libraries

- [React Use](#)
 - [Local Storage](#)
 - [Session Storage](#)

Milestone Breakdown



Milestone 7

Deliverables

- **Sharing**

Allow the user to share products with friends over social media platforms. Ensure that a link back to the app is included. Optionally allow users to download a product's details as a PDF.

- **Deep Linking**

Open your app when a user clicks on a link on their device.

Resources

- **BRS Requirements**

None

- **Designs**

Figma → Ready for Development → Home → [Product Enhancement](#)

Useful Links

Deep Linking

- [React Router](#)

PDFs with React

- [React PDF](#)

Milestone Breakdown

Milestone 6

Add to Cart
Data Persistence

Milestone 7

Sharing
Deep Linking

Milestone 8

App Environment Variants
Building for Production

Milestone 8

Deliverables

- **App Environment Variants**

Allow the app to be built for multiple environments. Change the colour scheme per environment. Allow features to be disabled for certain environments.

- **Building for Production**

Optimise your web app for production. Implement best practices such as source-map deletion, obfuscation, minification/uglification, and tree-shaking to improve the security of your production app.

Resources

- **BRS Requirement**

None

- Designs

Figma → [Colours](#)

Useful Links

Env Variables

- [Vite Env Variables](#)

Building for Production

- [Vite Production Building](#)
- [Vite Build Options](#)
- [Vite Obfuscator](#)

Additional Resources

Learn React

- [Quick Start](#)
- [Tutorial: Tic-Tac-Toe](#)
- [Thinking in React](#)
- [Describing the UI](#)
- [Adding Interactivity](#)
- [Managing State](#)

Useful React Libraries

- Component Libraries:
 - [Tailwind CSS](#)
 - [React Semantic UI](#)
 - [React Flowbite](#)
- Form Libraries:
 - [React Hook Form](#)
 - [Formik](#)
- Routing Library:
 - [React Router Tutorial](#)
- Request Libraries:
 - [Axios](#)
 - [React Query](#)
- File Picker Libraries:
 - [Use File Picker](#)
 - [Upload Widget React](#)
- PDF Library:
 - [React PDF](#)
- Useful Hook Library:
 - [React Use](#)

Vite

- [Scaffolding Your First Project](#)
- [Env Variables](#)
- [Production Building](#)
- [Build Options](#)
- [Obfuscator](#)

React State & Hooks

- [useState](#)
- [useEffect](#)
- [useCallback](#)
- [useMemo](#)
- [Custom Hooks](#)
- [React Context](#)

React Testing

- [React Testing Library](#)
- [Jest](#)

Firebase

- [Firebase For Web](#)
- [Firebase Analytics For Web](#)

Data Persistence

- [Using Local and Session storage in React](#)
- [Browser Local Storage](#)